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In[1]:= (*Задание 1*)
f[x_] = 4 (Cos[x])^3;
X = {0,  $\pi/6$ ,  $\pi/4$ ,  $\pi/3$ ,  $\pi/2$ ,  $\pi$ ,  $9\pi/5$ }; xx =  $\pi/7$ ;
F = f[X];
xk_ := X[[k+1]];
fk_ := F[[k+1]];
NN = Length[X];
n =  $\frac{NN-1}{2}$ ;
koef =
Solve[Table[a0 +  $\sum_{k=1}^n (a_k \sin[k * x_j] + b_k \cos[k * x_j]) = f_j$ , {j, 0, NN-1}], {}] // Flatten;

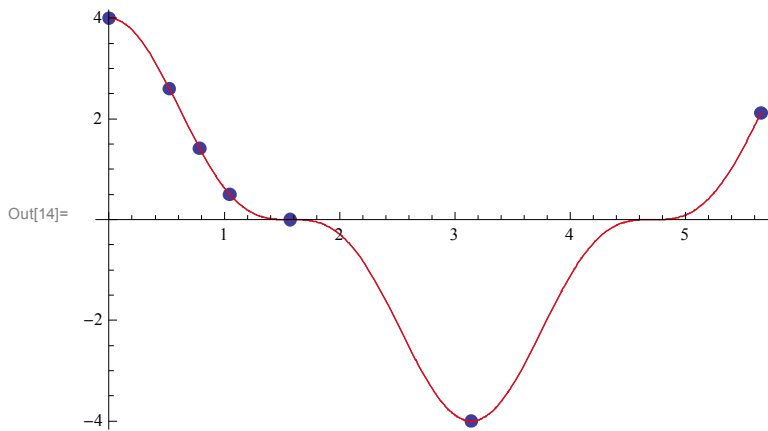
T[x_] = a0 +  $\sum_{k=1}^n (a_k \sin[k * x] + b_k \cos[k * x])$  /. koef // Simplify
Table[(T[xj] // N) == (fj // N), {j, 0, NN-1}]

Gr1 = ListPlot[MapThread[List, {X, F}], PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[f[y], {y, x0, xNN-1}];
Gr3 = Plot[T[y], {y, x0, xNN-1}, PlotStyle -> Red];
Show[Gr1, Gr2, Gr3]
T[xx] // N
Abs[f[xx] - T[xx]]

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Out[9]= $4 \cos[x]^3$

Out[10]= {True, True, True, True, True, True, True}



Out[15]= 2.92543

Out[16]= 0

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In[17]:= (*Задание 2*)
f[x_] = 4 Cos[x]^3;
NN = 7;

X = Table[x_k =  $\frac{2 \pi * k}{NN}$ , {k, 0, NN - 1}];

F = f[X];
f_k_ := F[[k + 1]];

n =  $\frac{NN - 1}{2}$ ;

Do[C_k =  $\frac{1}{NN} \sum_{j=0}^{NN-1} f_j * \text{Exp}[i * \frac{2 \pi}{NN}]^{-k*j}$ , {k, -n, n}];

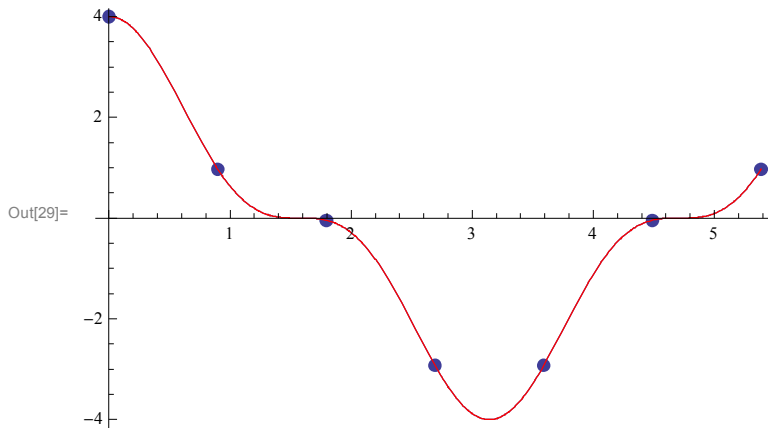
T[x_] =  $\sum_{k=-n}^n C_k * \text{Exp}[i * x]^k$  // FullSimplify
Table[(T[x_j] // N) == (f_j // N), {j, 0, NN - 1}]

Gr1 = ListPlot[MapThread[List, {X, F}], PlotStyle -> {PointSize[0.02]}];
Gr2 = Plot[f[y], {y, x_0, x_{NN-1}}];
Gr3 = Plot[T[y], {y, x_0, x_{NN-1}}, PlotStyle -> Red];
Show[Gr1, Gr2, Gr3, PlotRange -> All]

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Out[24]= $4 \cos[x]^3$

Out[25]= {True, True, True, True, True, True, True}



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In[30]:= (*Задание 3*)
XX = {1/10, 2/10, 3/10, 4/10}; f = {2/14, 3/15, 4/16, 5/17};
n = Length[XX] - 1; xx_k_ := XX[[k + 1]]; f_k_ := f[[k + 1]]

eqv[m_, y_] := Table[(D[x^i, {x, m}] /. x -> y) ==  $\sum_{k=0}^n d_k * xx_k^i$ , {i, 0, n}]

Pr[m_, y_] :=  $\sum_{k=0}^n d_k * f_k$  /. (Solve[eqv[m, y], {}]) // Flatten

{Pr[1, 1/10], Pr[1, 2/10], Pr[2, 0], Pr[3, 5/10]}

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Out[34]= $\left\{ \frac{291}{476}, \frac{127}{238}, -\frac{115}{119}, \frac{150}{119} \right\}$

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In[35]:= P[x_] = InterpolatingPolynomial[Table[{xx_k, f_k}, {k, 0, n}], x];
Pr1[m_, y_] := D[P[x], {x, m}] /. x -> y
{Pr1[1, 1/10] == Pr[1, 1/10], Pr1[1, 2/10] == Pr[1, 2/10],
 Pr1[2, 0] == Pr[2, 0], Pr1[3, 5/10] == Pr[3, 5/10]}
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Out[37]= {True, True, True, True}
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In[38]:= Pogr[m_, y_] := 
$$\frac{M}{(n+1)!} \text{Abs}\left[D\left[\prod_{k=0}^n (x - xx_k), \{x, m\}\right] /. x \rightarrow y\right]$$

{Pogr[1, 1/10], Pogr[1, 2/10], Pogr[2, 0], Pogr[3, 5/10]}
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Out[39]=  $\left\{\frac{M}{4000}, \frac{M}{12000}, \frac{7M}{240}, \frac{M}{4}\right\}$ 
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In[40]:= Table[

$$\frac{M}{(n+1)!} \text{Maximize}\left[\left\{\text{Abs}\left[D\left[\prod_{k=0}^n (x - xx_k), \{x, m\}\right]\right\}, 0 \leq x \leq 5/10\right], x][[1]], \{m, 0, n+2\}]$$

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Out[40]=  $\left\{\frac{M}{10000}, \frac{M}{480}, \frac{7M}{240}, \frac{M}{4}, M, 0\right\}$ 
```